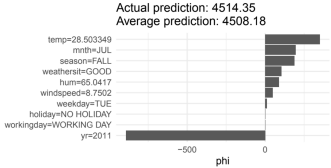
# Interpretable Machine Learning

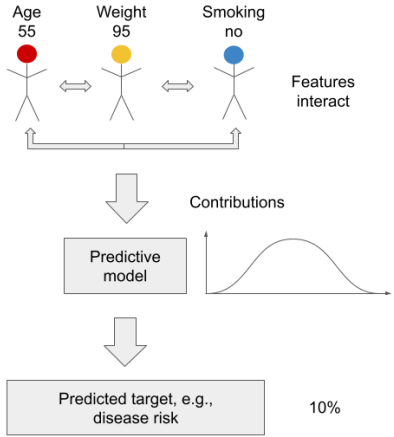
# Shapley Values for Local Explanations

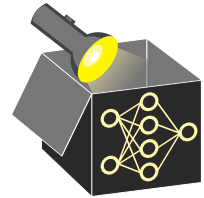



## Learning goals

- See model predictions as a cooperative game
- Transfer the Shapley value concept from game theory to machine learning

# FROM GAME THEORY TO MACHINE LEARNING



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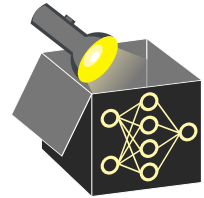

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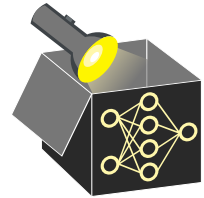

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- Players: Features $x_j, j \in \{1, \ldots, p\}$ which cooperate to produce a prediction
  $\rightsquigarrow$ How can we make a prediction with a subset of features without changing the model?
  $\rightsquigarrow$ PD function: $\hat{f}_S(\mathbf{x}_S) := \int_{X_{-S}} \hat{f}(\mathbf{x}_S, X_{-S}) d\mathbb{P}_{X_{-S}}$ ("removing" by marginalizing over $-S$)

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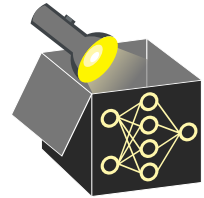

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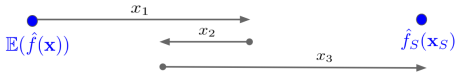
- Value function / payout of coalition $S \subseteq P$ for observation $\mathbf{x}$:

$$v(S) = \hat{f}_S(\mathbf{x}_S) - \mathbb{E}_{\mathbf{x}}(\hat{f}(\mathbf{x})), \text{ where } \hat{f}_S : \mathcal{X}_S \mapsto \mathcal{Y}$$

$\rightsquigarrow$ subtraction of $\mathbb{E}_{\mathbf{x}}(\hat{f}(\mathbf{x}))$ ensures that $v$ is a value function with $v(\emptyset) = 0$



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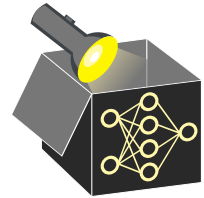

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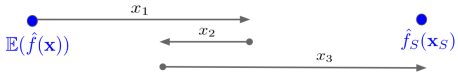
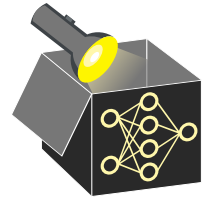



- Marginal contribution: $v(S \cup \{j\}) - v(S) = \hat{f}_{S \cup \{j\}}(\mathbf{x}_{S \cup \{j\}}) - \hat{f}_S(\mathbf{x}_S)$
  $\rightsquigarrow$ $\mathbb{E}_{\mathbf{x}}(\hat{f}(\mathbf{x}))$ cancels out due to the subtraction of value functions

# SHAPLEY VALUE - DEFINITION

Shapley (1953) | Strumbelj et al. (2014)



Shapley value $\phi_j$ of feature $j$ for observation $\mathbf{x}$ via **order definition**:

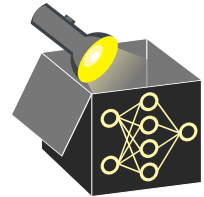
$$\phi_j(\mathbf{x}) = \frac{1}{|P|!} \sum_{\tau \in \Pi} \underbrace{\hat{f}_{S_j^\tau \cup \{j\}}(\mathbf{x}_{S_j^\tau \cup \{j\}}) - \hat{f}_{S_j^\tau}(\mathbf{x}_{S_j^\tau})}_{\text{marginal contribution of feature } j}$$

- Interpretation: Feature $x_j$ contributed $\phi_j$ to difference between $\hat{f}(\mathbf{x})$ and average prediction
  $\rightsquigarrow$ Note: Marginal contributions and Shapley values can be negative
- For exact computation of $\phi_j(\mathbf{x})$, the PD function $\hat{f}_S(\mathbf{x}_S) = \frac{1}{n}\sum_{i=1}^n \hat{f}(\mathbf{x}_S, \mathbf{x}_{-S}^{(i)})$ for any set of features $S$ can be used which yields

$$\phi_j(\mathbf{x}) = \frac{1}{|P|! \cdot n} \sum_{\tau \in \Pi} \sum_{i=1}^{n} \hat{f}(\mathbf{x}_{S_j^\tau \cup \{j\}}, \mathbf{x}_{-\{S_j^\tau \cup \{j\}\}}^{(i)}) - \hat{f}(\mathbf{x}_{S_j^\tau}, \mathbf{x}_{-S_j^\tau}^{(i)})$$

$\rightsquigarrow$ Note: $\hat{f}_S$ marginalizes over all other features $-S$ using all observations $i = 1, \ldots, n$

# ESTIMATION: A PRACTICAL PROBLEM



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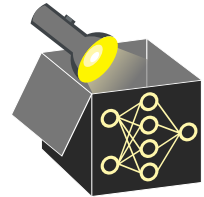

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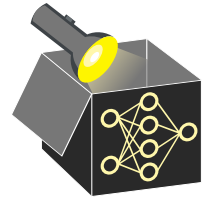

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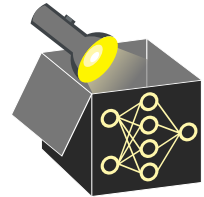

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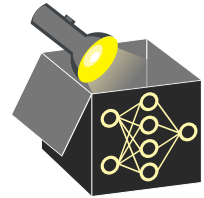
- Solution to both problems is sampling: Instead of averaging over $|P|! \cdot n$ terms, we approximate it using a limited amount of $M$ random samples of $\tau$ to build coalitions $S_j^\tau$
- $M$ is a tradeoff between accuracy of the Shapley value and computational costs
  $\rightsquigarrow$ The higher $M$, the closer to the exact Shapley values, but the more costly the computation

# APPROXIMATION ALGORITHM

Strumbelj et al. (2014)



Estimation of $\phi_j$ for observation $\mathbf{x}$ of model $\hat{f}$ fitted on data $\mathcal{D}$ using sample size $M$:

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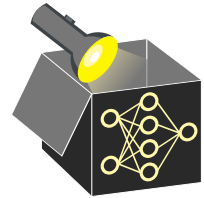

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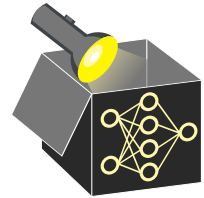

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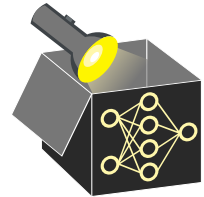

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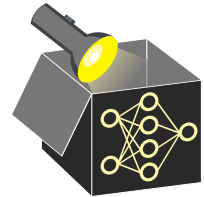

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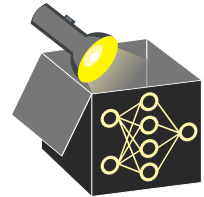

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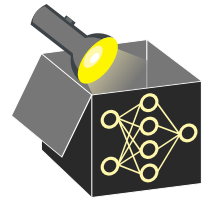

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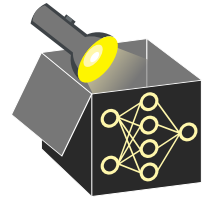

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   5. Compute difference $\phi_j^m = \hat{f}(\mathbf{x}_{+j}^{(m)}) - \hat{f}(\mathbf{x}_{-j}^{(m)})$
      $\rightsquigarrow \hat{f}_{S_m}(\mathbf{x}_{S_m})$ is approximated by $\hat{f}(\mathbf{x}_{-j}^{(m)})$ and $\hat{f}_{S_m \cup \{j\}}(\mathbf{x}_{S_m \cup \{j\}})$ by $\hat{f}(\mathbf{x}_{+j}^{(m)})$ over $M$ iters.

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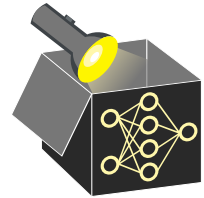

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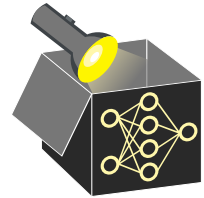
2. Compute Shapley value $\phi_j = \frac{1}{M} \sum_{m=1}^{M} \phi_j^m$

# SHAPLEY VALUE APPROXIMATION - ILLUSTRATION



**Definition**

$\mathbf{x}$: obs. of interest

$\mathbf{x}$ with feature values in $S_m$ (other are replaced)

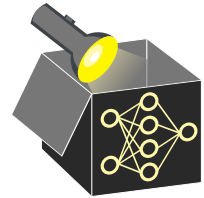
$$\phi_j(\mathbf{x}) = \frac{1}{M}\sum_{m=1}^{M}\left[\hat{f}(\mathbf{x}_{+j}{}^{(m)}) - \hat{f}(\mathbf{x}_{-j}{}^{(m)})\right]$$

$\mathbf{x}$ with feature values in $S_m \cup \{j\}$

| | Temperature | Humidity | Windspeed | Year |
|---|---|---|---|---|
| $x$ | 10.66 | 56 | 11 | 2012 |
| $x_{+j}$ | 10.66 | 56 | $random: z_{windspeed}^{(m)}$ | 2012 |
| $x_{-j}$ | 10.66 | 56 | $random: z_{windspeed}^{(m)}$ | $random: z_{year}^{(m)}$ |

$j$

# SHAPLEY VALUE APPROXIMATION - ILLUSTRATION



**Definition**

Contribution of feature $j$ to coalition $S_m$

$$\phi_j(\mathbf{x}) = \frac{1}{M}\sum_{m=1}^{M}\underbrace{\left[\hat{f}(\mathbf{x}_{+j}{}^{(m)}) - \hat{f}(\mathbf{x}_{-j}{}^{(m)})\right]}_{:=\Delta(j,S_m)}$$

- $\Delta(j, S_m) = \hat{f}(\mathbf{x}_{+j}^{(m)}) - \hat{f}(\mathbf{x}_{-j}^{(m)})$ is the marginal contribution of feature $j$ to coalition $S_m$
- Here: Feature *year* contributes +700 bike rentals if it joins coalition $S_m = \{temp, hum\}$

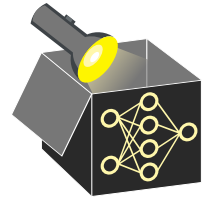
| | Temperature | Humidity | Windspeed | Year | Count |
|---|---|---|---|---|---|
| $x$ | 10.66 | 56 | 11 | 2012 | |
| $x_{+j}$ | 10.66 | 56 | $random: z_{windspeed}^{(m)}$ | 2012 | 5600 |
| $x_{-j}$ | 10.66 | 56 | $random: z_{windspeed}^{(m)}$ | $random: z_{year}^{(m)}$ | 4900 |

$j$ $\hat{f}$

700

$\Delta(j, S_m)$ marginal contribution

# SHAPLEY VALUE APPROXIMATION - ILLUSTRATION

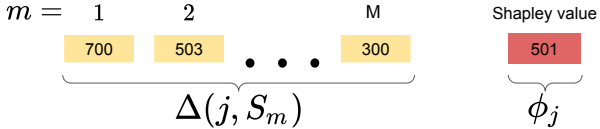


**Definition**

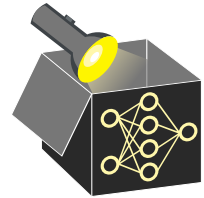
average the contributions of feature $j$

$$\phi_j(\mathbf{x}) = \frac{1}{M}\sum_{m=1}^{M}\left[\hat{f}(\mathbf{x}_{+j}{}^{(m)}) - \hat{f}(\mathbf{x}_{-j}{}^{(m)})\right]$$

- Compute marginal contribution of feature $j$ towards the prediction across all randomly drawn feature coalitions $S_1, \ldots, S_m$
- Average all $M$ marginal contributions of feature $j$
- Shapley value $\phi_j$ is the payout of feature $j$, i.e., how much feature *year* contributed to the overall prediction in bicycle counts of a specific observation $\mathbf{x}$



# REVISITED: AXIOMS FOR FAIR ATTRIBUTIONS



We take the general axioms for Shapley Values and apply it to predictions:

- **Efficiency**: Shapley values add up to the (centered) prediction: $\sum_{j=1}^{p} \phi_j = \hat{f}(\mathbf{x}) - \mathbb{E}_{\mathbf{x}}(\hat{f}(X))$

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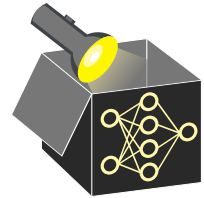

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- **Symmetry**: Two features $j$ and $k$ that contribute the same to the prediction get the same payout
  $\rightsquigarrow$ interaction effects between features are fairly divided
  $\hat{f}_{S\cup\{j\}}(\mathbf{x}_{S\cup\{j\}}) = \hat{f}_{S\cup\{k\}}(\mathbf{x}_{S\cup\{k\}})$ for all $S \subseteq P \setminus \{j, k\}$ then $\phi_j = \phi_k$

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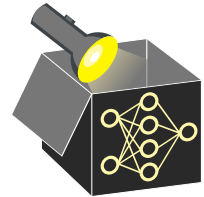

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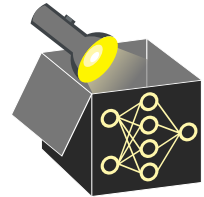

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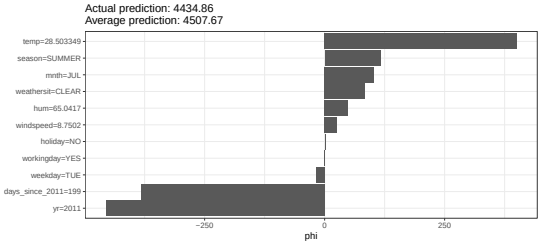
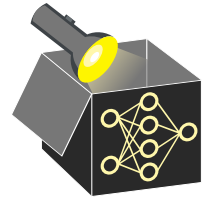
- **Additivity**: For a prediction with combined payouts, the payout is the sum of payouts: $\phi_j(v_1) + \phi_j(v_2) \rightsquigarrow$ Shapley values for model ensembles can be combined

# BIKE SHARING DATASET




- Shapley values of observation $i = 200$ from the bike sharing data
- Difference between model prediction of this observation and the average prediction of the data is fairly distributed among the features (i.e., $4434 - 4507 \approx -73$)
- Feature value temp = 28.5 has the most positive effect, with a contribution (increase of prediction) of about +400

# ADVANTAGES AND DISADVANTAGES



**Advantages:**

- **Solid theoretical foundation** in game theory
- Prediction is **fairly distributed** among the feature values $\rightsquigarrow$ easy to interpret for a user
- **Contrastive explanations** that compare the prediction with the average prediction

**Disadvantages:**

- Without sampling, Shapley values need a lot of computing time to inspect all possible coalitions
- Like many other IML methods, Shapley values suffer from the inclusion of unrealistic data observations when features are correlated